

Author declarations

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CRedit authorship contribution statement: Jacob T. Dingwall: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Software, Validation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of competing interest: The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. For the required Elsevier declarations-tool upload, select “I have nothing to declare” and upload the generated Word document.

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Ethics and permits: Not applicable; the study used secondary observational datasets and involved no animal handling or human-participant research.

Data availability: DFO-derived source data are available through the Government of Canada Open Government Portal. eBird checklist data require authorized access under eBird terms and are not redistributed. Raw EBD/SED, protected row-level derivatives, observer identities, exact localities, exact coordinates, event tokens, and transformation mappings cannot be shared because of source terms and privacy controls. Privacy-safe aggregate tables, claim audits, provenance records, and publication figures are deposited at <https://github.com/JTDingwall/herring-x-ebird-version-2>. Authorized researchers can reconstruct the analysis from permitted source data, frozen registries, and the tagged code.

Code availability: Code, frozen specifications, privacy-safe aggregate outputs, and reproducibility instructions are available at <https://github.com/JTDingwall/herring-x-ebird-version-2>. The exact analysis freeze is commit `c54b8e7f95a2fe3573e2e38633079cd223c5a783`, tag `stage4a-publication-v2-analysis-freeze`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process: [AUTHOR TO CONFIRM AFTER REVIEW: During preparation of this work, the author used OpenAI Codex to assist with journal-specific organization, language

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